



KEMENTERIAN PENDIDIKAN, KEBUDAYAAN,
RISET, DAN TEKNOLOGI

Kampus
Merdeka
INDONESIA JAYA



Certificate of Completion

is proudly presented to

Mochammad Lanang Afkaar Arrachmandhika

for successfully completing **Bangkit, specializing in Machine Learning.**

Bangkit is a Google-led academy designed to produce high-caliber technical talents for world-class Indonesian technology companies and startups.

July 21, 2021

William Florence

Education Program Lead,
Asia Pacific - Google





May 2, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed

Structuring Machine Learning Projects

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A blue ink signature of Andrew Ng.

Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera
Kian Katanforoosh, Co-founder, Workera
Younes Bensouda Mourri, Instructor of AI, Stanford University

**COURSE
CERTIFICATE**



Verify at coursera.org/verify/QG57KVV2BBZ7

Coursera has confirmed the identity of this individual and their participation in the course.



4 Courses

Browser-based Models with
TensorFlow.js

Device-based Models with
TensorFlow Lite

Data Pipelines with
TensorFlow Data Services

Advanced Deployment
Scenarios with TensorFlow

May 30, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed the online, non-credit Specialization

TensorFlow: Data and Deployment

In this specialization, you continued to develop your understanding of machine learning with TensorFlow: Data and Deployment. You have gone beyond basic modeling and learned how to train and run your models within a browser, optimize machine learning models for mobile devices, and create effective data pipelines with TensorFlow Data Services. Now that you've learned the various ways to deploy your models, you're well-prepared to take your models into the hands of real people on all kinds of devices!

Laurence Moroney

Laurence Moroney
Lead AI Advocate at
Google

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:
coursera.org/verify/specialization/D2L4DJSDQQX4



4 Courses

Introduction to TensorFlow
for Artificial Intelligence,
Machine Learning, and Deep
Learning

Convolutional Neural
Networks in TensorFlow

Natural Language
Processing in TensorFlow

Sequences, Time Series and
Prediction

Apr 24, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed the online, non-credit Professional
Certificate

DeepLearning.AI TensorFlow Developer

Congratulations! You have completed all 4 courses of the DeepLearning.AI TensorFlow Developer Professional Certificate program. As part of this Professional Certificate program, you have learned: how to build and train neural networks using TensorFlow, how to improve network performance using convolutions as you train it to identify real-world images, how to teach machines to understand, analyze, and respond to human speech with natural language processing systems, and more! These, and other TensorFlow concepts, are going to be at the forefront of the coming transformation to an AI-powered future.

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

John Ng Laurence Moroney

Laurence Moroney
Lead AI Advocate
Google

Andrew Ng
Founder
DeepLearning.AI

Verify this certificate at:
[coursera.org/verify/professional-
cert/F6D8RNMMLVQV3](https://coursera.org/verify/professional-cert/F6D8RNMMLVQV3)



Mar 15, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed the online, non-credit Professional Certificate

Google IT Automation with Python

This six-course certificate, developed by Google, is designed to provide IT professionals with in-demand skills -- including Python, Git, and IT automation -- that can help them advance their careers. The hands-on curriculum is designed to teach learners how to write code in Python, with a special focus on how this applies to automating tasks in the world of IT support and systems administration. Those who received this certificate passed all graded assessments with a score of 80% or above. They should have a strong foundation in how to use Git and GitHub, troubleshoot and debug complex problems, and apply automation at scale by using configuration management and the Cloud in order to prepare them for more advanced IT Support Specialist or Junior Systems Administrator positions.

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:
[coursera.org/verify/professional-
cert/95NGNQ5RC626F](https://coursera.org/verify/professional-cert/95NGNQ5RC626F)



6 Courses

- Crash Course on Python
- Using Python to Interact with the Operating System
- Introduction to Git and GitHub
- Troubleshooting and Debugging Techniques
- Configuration Management and the Cloud
- Automating Real-World Tasks with Python



Mar 19, 2021

MOCH.LANANG AFKAAR ARACHMANDIK

has successfully completed the online, non-credit Professional Certificate

Google IT Support

Those who earn the Google IT Support Professional Certificate have completed five courses, developed by Google, that include hands-on, practice-based assessments and are designed to prepare them for entry-level roles in IT support. They are competent in foundational skills, including troubleshooting and customer service, networking, operating systems, system administration, and security.

5 Courses



Technical Support
Fundamentals

The Bits and Bytes of
Computer Networking

Operating Systems and You:
Becoming a Power User

System Administration and
IT Infrastructure Services

IT Security: Defense against
the digital dark arts

Google

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Verify this certificate at:
coursera.org/verify/professional-cert/4775PV9X38NJ



3 Courses

Mathematics for Machine
Learning: Linear Algebra
Mathematics for Machine
Learning: Multivariate
Calculus
Mathematics for Machine
Learning: PCA

Imperial College
London

Mar 28, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed the online, non-credit Specialization

Mathematics for Machine Learning

A sequence of 3 courses on the prerequisite mathematics for applications in data science and machine learning. Successful participants learn how to represent data in a linear algebra context and manipulate these objects mathematically. They are able to summarise properties of data sets and map them onto lower dimensional spaces with principal component analysis. Finally they can solve optimisation problems and use this skill to train models for describing data such as simple neural networks.

David Dye
David Dye

David Dye
Professor of Metallurgy
Department of
Materials
Imperial College London

Samuel J. Cooper
Lecturer

Dyson School of Design
Engineering
Imperial College London

Marc Deisenroth
Senior Lecturer
Department of
Computing
Imperial College London

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Verify this certificate at:
coursera.org/verify/specialization/S5DJP9SCSNSY

**Imperial College
London**

Mar 11, 2021

**MOCH.LANANG AFKAAR
ARACHMANDIK**

has successfully completed

Mathematics for Machine Learning: Linear Algebra

an online non-credit course authorized by Imperial College London and offered through
Coursera



David Dye, Samuel J. Cooper, and A. Freddie Page

**COURSE
CERTIFICATE**



Verify at coursera.org/verify/73RMQEDZ7AWG

Coursera has confirmed the identity of this individual and their participation in the course.



CERTIFICATE OF ACHIEVEMENT

This is to certify that

MOCHAMMAD LANANG AFKAAR AR.

achieved the following scores on the

TOEFL ITP® Test

Listening Comprehension:	52
Structure & Written Expression:	49
Reading Comprehension:	61
Total:	540

Under the auspices of:
LANGUAGE CENTER TELKOM UNIVERSITY
At: BANDUNG, Indonesia
Date: 04 Dec 2020
Bronze: 08899507

Mohammad Kousha

Mohammad Kousha
General Manager
Global Education Division, ETS